

This Zenodo deposit contains a publicly available description of the Dataset:

"Bulk RNA-seq analysis of G2019S LRRK2 knockin primary astrocytes".

Dataset Description:

Primary astrocytes were isolated from P1 G2019S LRRK2 knockin pups and their wildtype littermates. The cells were treated either with Etoposide(20uM) or vehicle at 18 days in vitro; cells were collected at 24 days in vitro (n=3/group). Total RNA was isolated from cell pellets, then subjected to bulk RNA-seq.

Authors:

- Ma, Yue; [ORCID:0000-0002-1031-5175](https://orcid.org/0000-0002-1031-5175); Van Andel Institute
- Moore, Darren; [ORCID:0000-0003-2359-5679](https://orcid.org/0000-0003-2359-5679); Van Andel Institute

ASAP Team: Lee

Dataset Name: lee-mouse-bulk-rnaseq-astro-g2019s-lrrk2, v1.0

Principal Investigator: Darren Moore Darren.moore@vai.org

Dataset Submitter: Yue Ma Yue.ma@vai.org

Publication DOI:

Grant IDs: ['ASAP-000592']

ASAP Lab:

ASAP Project: Are there bidirectional pathological interactions between senescence and LRRK2 models of PD

Project Description: Mutations in LRRK2 gene cause late-onset, autosomal dominant forms of familial PD that are clinically indistinguishable from sporadic PD. In addition, LRRK2 gene polymorphisms are associated with sporadic PD risk. The late disease-onset with these PD-linked genes suggest that interactions with age-related pathways are critical for manifesting or unmasking their pathogenic actions. Studies in experimental models further suggest that PD-linked LRRK2 mutations can induce abnormal mitochondrial function, lysosomal activity, autophagy, proteostasis and inflammatory processes, pathways that are dysregulated by cellular senescence. Our project focuses on determining whether:(1) PD-linked mutations in LRRK2 induce cellular senescence in the rodent brain, (2) increased senescence can exacerbate or unmask PD-like neurodegenerative phenotypes in LRRK2 mouse models, and (3) if removing senescent cells will provide neuroprotection in LRRK2 rodent models of PD.

Submission Date: 2026-09-22

This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

This research was funded by the Aligning Science Across Parkinson's Collaborative Research Network (ASAP CRN), through the Michael J. Fox Foundation for Parkinson's Research (MJFF).

This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset